

Detailed Marking Report — Mathematics 2nd ATTEMPT

exam-11-ch1-2-4-5-7 (Version 2) | Units 1, 2, 4, 5, 7 | Federal Board Annual Examination 2026
Candidate: Seemab | Attempt date: 5 July 2026 | Marked: 5 July 2026 | Re-sit of the same paper first attempted 3 July 2026 (58.5/65)

63.5/65	97.7%	A+	12/12	32/33	19.5/20
TOTAL SCORE	PERCENTAGE	GRADE	SECTION A (MCQ)	SECTION B	SECTION C

All three sections present and clearly readable across 15 handwritten pages (scanned via CamScanner). Seemab answered Section C first (pages 2-8), then Section B (pages 8-15), then listed all 12 MCQ answers on page 1, which is allowed. All 11 Section B questions and all 4 Section C questions were attempted — nothing left blank this time. This is a re-attempt of exam-11-ch1-2-4-5-7 (Version 2), the exact same paper Seemab sat on 3 July 2026 (scored 58.5/65, 90.0%). Every deduction below was checked against the exam paper's own worked solution and reconciled so the deduction log total (1.5 marks lost) agrees with the headline score (63.5/65).

Section A — Multiple Choice Questions (12 × 1 = 12 Marks)

MCQ answers listed on page 1. Every letter matches the Version 2 key exactly — a clean 12/12, identical to the 3 July attempt.

#	Topic	Key	Seemab	Result	Mark
1	Quotient rule of exponents	B	B	✓	1
2	$\sqrt{a}$ for $a > 0$ is non-negative real	C	C	✓	1
3	Scientific notation of 0.000307	B	B	✓	1
4	$\log_3 81$	C	C	✓	1
5	$\log 18$ via $\log 2$ and $\log 3$	B	B	✓	1
6	Factorization of $x^2 - 9x + 20$	A	A	✓	1
7	$\text{HCF} \times \text{LCM} = P \times Q$	C	C	✓	1
8	Standard form of a linear equation	B	B	✓	1
9	Solution set of $ 3x - 6 = 9$	A	A	✓	1
10	Distance (0,0) to (5,12)	C	C	✓	1
11	Midpoint of A(4, -2), B(-2, 8)	B	B	✓	1
12	HCF of $x^2 - 4$ and $x^2 - x - 2$	B	B	✓	1

Section A Total	12 / 12
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Perfect 12/12, same as 3 July. This extends the current consecutive-perfect-MCQ streak to 12 exams. Math MCQ all-time: 83/84.

Section B — Short Questions (11 × 3 = 33 Marks)

Seemab answered the OR variant for Q2(i), Q2(iii); the main for all others — identical choice pattern to 3 July. Ten of eleven questions scored full marks.

Q2(i) OR — Equivalent radicals: (i) $(-8)^{1/3}$ (ii) $(\sqrt{25})^4$ (iii) $(x^{-3})^2 \cdot (x^0)^5$ **3/3**

(i) $(-8)^{1/3} = ((-2)^3)^{1/3} = -2$. *Correct (1/1)* — *Watch:* an unexplained "or $1/2$ " was appended right after this fully-derived correct value. No marks lost since -2 was reached through valid steps first, but writing two competing final answers is a habit to drop — commit to one boxed answer.

(ii) $(\sqrt{25})^4 = 5^4 = 625$. *Correct (1/1)*

(iii) $(x^{-3})^2 \cdot (x^0)^5 = x^{-6} \cdot 1 = x^{-6} (1/x^6)$. *Correct (1/1)*

Q2(ii) main — True/False with counterexample**3/3**

(a) Every integer is rational — TRUE. *Correct (1/1)*

(b) Every irrational is real — TRUE. *Correct (1/1)*

(c) Every rational is an integer — FALSE, counterexample: $1/2$ is rational but not an integer. *Correct (1/1)* — *Improvement:* the counterexample was missing on 3 July (cost 0.5 mark then); it is written in full this time.

Q2(iii) OR — Notation conversion**3/3**

(i) $7.05 \times 10^{-4} = 0.000705$. *Correct (1/1)*

(ii) $2.31 \times 10^6 = 2,310,000$. *Correct (1/1)*

(iii) $0.0000000512 = 5.12 \times 10^{-8}$. *Correct (1/1)*

Q2(iv) main — Find x **2/3**

(i) $\log_x 125 = 3 \Rightarrow x^3 = 125 = 5^3 \Rightarrow x = 5$. *Correct (1/1)*

(ii) $\log_4 x = -3/2 \Rightarrow x = 2^{-3}$. *-0.5:* evaluated 2^{-3} incorrectly as -8 first (negative-exponent/sign confusion — a negative exponent means reciprocal, not a negative number), then separately wrote "or $x = 1/8$ " beside it. The correct value $1/8$ is present but the contradicting -8 shows the rule wasn't confidently applied. (0.5/1)

(iii) $\log_2(x^2 - 3) = \log_2 13 \Rightarrow x^2 = 16 \Rightarrow x = 4$. *-0.5:* only the positive root shown; the key answer is $x = \pm 4$ (both satisfy the domain $x^2 - 3 > 0$). This is the exact same omission on this identical sub-question in the 3 July attempt — a repeat, not a new slip. (0.5/1)

Q2(v) main — Log using tables**3/3**

(i) $\log 4580 = 3.6609$. *Correct (1/1)*

(ii) $\log 0.00093 = 4.9685$. *Correct (1/1)*

(iii) $\log 54 = 1.7324$. *Correct (1/1)*

Q2(vi) main — Factorize completely**3/3**

(i) $x^4 + 4m^4 = (x^2 + 2m^2 + 2mx)(x^2 + 2m^2 - 2mx)$. *Correct (1.5/1.5)*

(ii) $3x^2 + 11x + 6 = (3x + 2)(x + 3)$. *Correct (1.5/1.5)*

Q2(vii) main — HCF by factorization**3/3**

(i) $x^2 - 49 = (x + 7)(x - 7)$. *Correct (1/1)*

(ii) $x^2 - 4x - 21 = (x - 7)(x + 3)$. *Correct (1/1)* — *Improvement:* the 3 July attempt mis-copied this exact polynomial's constant as $+21$ and factored it wrong; it is fully correct this time.

HCF = $(x - 7)$. *Correct (1/1)*

Q2(viii) main — Solve $(2x+3)/5 = (3-4x)/8$ **3/3**

Cross-multiplied $8(2x+3) = 5(3-4x) \Rightarrow 36x = -9 \Rightarrow x = -1/4$. *Correct (3/3)*

Q2(ix) main — Solve $|4x-3| = 13$ **3/3**

Both cases solved: $x = 4$ and $x = -5/2$. Solution set $\{4, -5/2\}$. *Correct (3/3)*

Q2(x) main — Collinearity of A(1,-2), B(7,8), C(4,3)**3/3**

$AB = 2\sqrt{34}$, $BC = \sqrt{34}$, $AC = \sqrt{34}$; $BC + AC = AB \Rightarrow$ points collinear. *Correct (3/3)*

Q2(xi) main — Midpoint application**3/3**

$P(3,k)$ midpoint of $A(-1,7)$, $B(7,-3) \Rightarrow k = 2$; then $PA = \sqrt{41}$ units. *Correct (3/3)*

Section B Total**32 / 33****Section C — Long Questions (4 × 5 = 20 Marks)**

Seemab answered the OR variant for Q3(i) and Q3(ii); the main for Q3(iii) and Q3(iv) — identical choice pattern to 3 July. This is where almost every mark from the 3 July attempt was recovered.

Q3(i) OR — Evaluate $x = {}^3\sqrt{(8.59) \times (55.6)^2 \div (2.51 \times \sqrt{2.12})}$ using log tables**5/5**

Expansion $\log x = (1/3)\log 8.59 + 2 \log 55.6 - \log 2.51 - (1/2)\log 2.12$ set up correctly. Table values used (0.9339, 1.7450, 0.3996, 0.3263) all match the key within normal table-rounding tolerance. Combined cleanly: $0.3113 + 3.4900 - 0.3996 - 0.1632 = 3.23855$ (key: 3.2386) — no arithmetic jump this time. Final **antilog** ≈ 1732 is correct. *Correct throughout (5/5)*

Improvement: the 3 July attempt had an internally inconsistent intermediate line (3.56545) despite still landing on the right antilog; this attempt's working is consistent end to end.

Q3(ii) OR — (a) $\sqrt{4a^2+12ab+9b^2}$ (b)(i) $(x^2+3x-4)(x^2+3x+5)+8$ (b)(ii) $8a^3-125b^3-2a+5b$ **5/5**

(a) $4a^2+12ab+9b^2 = (2a+3b)^2 \Rightarrow \sqrt{} = \pm(2a+3b)$. *Correct (1.5/1.5)* — *Improvement:* the 3 July attempt stopped at the squared form and never took the root itself, costing 0.5 then; the $\pm$ root is written explicitly this time.

(b)(i) Let $u = x^2+3x$: $(u-4)(u+5)+8 = u^2+u-12 = (u+4)(u-3) = (x^2+3x+4)(x^2+3x-3)$. *Correct (2/2)*

(b)(ii) $(2a)^3-(5b)^3-(2a-5b) = (2a-5b)(4a^2+10ab+25b^2-1)$. *Correct (1.5/1.5)*

Q3(iii) main — (a) Solve $(y+1)/3 + (y+1)/2 = 2 - (y+3)/2$ (b) Solve $(3/4)x - 1 \geq x + 1$ in set notation**5/5**

(a) LHS combined correctly to $(5y+5)/6$. RHS correctly simplified this time: $2 - (y+3)/2 = (4-y-3)/2 = (1-y)/2$ — the exact sign step that was flipped in the 3 July attempt (which turned it into $(7-y)/2$ and gave $y = 2$). Cross-multiplied $6(1-y) = 2(5y+5) \Rightarrow 6-6y = 10y+10 \Rightarrow -4 = 16y \Rightarrow y = -1/4$. *Correct (3/3)*

(b) $3x-4 \geq 4x+4 \Rightarrow -x \geq 8 \Rightarrow x \leq -8$; solution set $\{x \mid x \in \mathbb{R} \wedge x \leq -8\}$. *Correct (2/2)*

Major turnaround: this whole question scored just 1/5 on 3 July (sign error in (a), (b) left entirely blank). Both faults are fixed here — full marks.

Q3(iv) main — A(0,0), B(6,0), C(6,4), D(0,4): (a) sides not all equal (b) diagonals + midpoints $\rightarrow$ rectangle**4.5/5**

(a) $AB=6$, $BC=4$, $CD=6$, $DA=4$; opposite sides equal but adjacent sides unequal $\Rightarrow$ not all sides equal. *Correct (2/2)*

(b) $AC = BD = \sqrt{52}$ (diagonals equal); midpoint of $AC =$ midpoint of $BD = (3,2)$ (bisect each other) — both correctly computed. *-0.5:* the concluding sentence "hence ABCD is a rectangle" was never written; the page moves straight to the next question after the midpoint numbers. (2.5/3)

Small regression: this exact conclusion sentence WAS written on 3 July and earned full marks then — worth reforming the habit of stating the geometric conclusion, not just the numbers.

Section C Total**19.5 / 20****Final Score Breakdown**

Section	Available	Earned	%	Notes
Section A — MCQ	12	12	100%	All 12 correct — 12th consecutive perfect MCQ score
Section B — Short Questions	33	32	97.0%	Lost 1.0: Q2(iv)(ii) sign confusion (0.5), Q2(iv)(iii) missing negative root (0.5)
Section C — Long Questions	20	19.5	97.5%	Lost 0.5: Q3(iv)(b) rectangle conclusion not stated
TOTAL	65	63.5	97.7%	Grade A+

Deduction Log (all 1.5 marks lost)

Question	Sub-part	Issue	Lost
Q2(iv)	(ii) $\log_4 x = -3/2$	2^{-3} miscalculated as -8 before self-correcting to $1/8$ — negative-exponent sign confusion.	-0.5
Q2(iv)	(iii) $x^2 = 16$	Gave only $x=4$; missed $x=-4$. Same omission repeated from the 3 July attempt on this identical question.	-0.5
Q3(iv)	(b) rectangle proof	Diagonals and midpoints correctly shown equal, but the concluding "ABCD is a rectangle" sentence was never written.	-0.5

Strengths in this attempt

- **Perfect MCQ (12/12)** for the 12th exam running — identical to 3 July, zero MCQ errors carried across.
- **Q3(iii) fully corrected:** the sign error in (a) and the blank inequality in (b) from 3 July are both fixed — jumped from $1/5$ to $5/5$, the single biggest swing in this re-attempt.
- **Q3(ii)(a) square root completed:** the $\pm(2a+3b)$ root is now written explicitly, closing the gap that cost 0.5 mark on 3 July.
- **Q3(i) log-table evaluation fully consistent** this time — correct antilog 1732 reached with no arithmetic inconsistency along the way.
- **Q2(vii) HCF factorization fully correct** — the mis-copied constant from 3 July (+21 instead of -21) is fixed.
- **Q2(ii) counterexample included** this time ($1/2$ is rational but not an integer) — was missing on 3 July.
- Absolute value equations, linear equations, and coordinate geometry (Q2(viii)–Q2(xi)) all clean and fully correct.

Areas to revise

- **Negative exponents (Q2(iv)(ii)):** 2^{-3} was briefly evaluated as -8 before self-correcting to $1/8$. A negative exponent means "take the reciprocal", it does not make the value negative. This is a new slip — it wasn't there in the 3 July attempt on the same sub-question.
- **Missing negative roots (Q2(iv)(iii)):** $x = \pm 4$ required, only $x = 4$ given — the exact same gap as 3 July on this identical question. Confirmed recurring pattern: whenever solving $x^2 = k$, state both $\pm\sqrt{k}$, then check the domain before dropping either root.
- **Q3(iv)(b) conclusion sentence dropped:** all the numeric work (equal diagonals, matching midpoints) was right, but the sentence "hence ABCD is a rectangle" was never written — and it WAS written correctly on 3 July. Always close a proof with the named conclusion, not just the numbers.
- **Watch (no marks lost):** Q2(i)(i) added an unexplained "or $1/2$ " after already correctly deriving -2 . Harmless this time, but two competing final answers in one line is a risky habit — commit to a single boxed answer.

Comparison vs Prior Attempt (same exam)

Attempt	Section A	Section B	Section C	Total	%	Grade
3 Jul 2026 (1st attempt)	12/12	31.5/33	15/20	58.5/65	90.0%	A+
5 Jul 2026 (2nd attempt, this report)	12/12	32/33	19.5/20	63.5/65	97.7%	A+
Change	steady	+0.5	+4.5	+5.0	+7.7 pts	—

Nearly all of the improvement came from Section C, where both of the 3 July errors (the Q3(iii)(a) sign flip and the blank Q3(iii)(b)) were corrected, and the Q3(ii)(a) square-root omission was closed. Section B held steady at essentially full marks, with one recurring gap (missing negative root in Q2(iv)(iii)) persisting from the first attempt, and one new but minor slip (Q2(iv)(ii) sign confusion) appearing this time.

Overall: A strong re-sit — 63.5/65 (97.7%, A+), up 5 marks from the 90.0% scored on 3 July. The two big Section C wounds from the first attempt are both healed. What's left is small and precise: remember both roots when solving $x^2=k$, take care with negative exponents, and always write the final concluding sentence on geometry proofs, not just the supporting numbers.